chapter twelve

CHI–SQUARE TESTS

Don't ask what it means,
but rather how it is used.

– L. Wittgenstein

INTRODUCTION

This chapter is an extension of the hypothesis testing we did with two proportions in chapter 10. We can use the Chi-Square distribution to test hypotheses that involve more than two population proportions. We will also use the Chi-Square distribution to test hypotheses about the population variance.

The test statistic used for proportions in this chapter involves what are called **observed frequencies** and **expected frequencies** that occur when we work with **categorical data**. A **contingency table** is used to record the observed frequencies in the **cells** of the table. The value of the test statistic is found by taking the sum of the squares of the difference between the observed and expected frequencies divided by the expected frequency for each category. The test statistic has an approximate chi-square distribution, and the critical value for a given level of significance is obtained from the chi-square table. Three of the tests of hypotheses that we will look at in this chapter are called a test for goodness of fit, a test of independence, and test for homogeneity.

The **test for goodness of fit** is one of the first hypothesis tests developed; it was first published in 1900 by Karl Pearson. This test can be illustrated using an example with the coins. In tossing a fair coin there are two categories, heads and tails. In ten tosses, we expect 5 heads and 5 tails, but we may only observe 4 heads and 6 tails. A hypothesis test of the equality of the proportions for heads and tails is done using the expected values of 5 and the observed values of 4 and 6 in the test statistic. We then compare it to a critical value obtained from the χ^2 table. Of course, the goodness of fit test can be extended to more categories.

The **test of independence** is used to determine if two or more characteristics of a single population are related. A typical example of a test of independence involves data on cholesterol levels for men and women to determine if cholesterol levels are independent of gender. The notation used to find the expected proportion in each cell is similar to what we used in the probability chapter when we had independent events. For example, the expected proportion of the sample that is female and has low cholesterol, p_{FL}, is found using the product $p_F \cdot p_L$. The observed and expected frequencies are then used in the test statistic, and it is compared to a critical value from the χ^2 table.

The **homogeneity test** is similar to the test of independence, but it involves more than one population. For example, we consider five different ethnic groups and measure the number of people in each group with high cholesterol. We test the hypothesis that the five population proportions are equal. The equality of the proportions is used to obtain the expected value for each category. The observed and expected frequencies are then used to get the value of the chi-square test statistic, and it is compared to the critical value from the χ^2 table.

Although the chi-square test statistic that is used in each of the above tests looks very different from the other test statistics we have used in hypothesis testing, we still use the same five steps to determine if we should reject the null hypothesis.

CHAPTER 11 HOT SPOTS

1. **z Test For Proportions Versus Chi-Square Test.**
   Starts on **page 11–4.** Mann 11.4

2. **Reporting P Values For The χ^2 Distribution**
   Starts on **page 11–5.** Problems on **pages 11–6, 11–7.**
   Mann 11.2, 11.4

If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic for a help session with your professor.

✷
Mann Section 11.4

Hot Spot #1 – z Test For Proportions Versus Chi-Square Test

Earlier we used a z test statistic to test a hypothesis of equal proportions. The chi-square test of homogeneity can also be used to test a hypothesis that two population proportions are equal. The chi-square homogeneity test has the advantage of being a more general hypothesis test that allows us to test whether two or more populations are homogeneous with respect to a characteristic. The following example shows that we get exactly the same results using the two different hypothesis tests on the same data. A survey of 50 men and 50 women asked them whether they liked country music. The results are entered into the table below:

Figure 11–1: Survey Of Country Music Tastes

Survey of Country Music Tastes

Respondents	No	Yes	Total
Men(1)	20	30	50
Women(2)	15	35	50
Totals	35	65	100

We want to test a null hypothesis that the population proportions are equal, against the alternate hypothesis that they are not equal, at $\alpha = 5\%$.

$$H_0\colon p_1 = p_2 \text{ versus } H_a\colon p_1 \neq p_2$$

The critical values for the two-tailed test are ±1.96, and we will test the hypothesis using the test statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\overline{p}\overline{q}\left[\frac{1}{n_1} + \frac{1}{n_2}\right]}}$$

$$\hat{p}_1 = \tfrac{30}{50} \qquad \hat{p}_2 = \tfrac{35}{50}$$

$$\overline{p} = \tfrac{30+35}{100} = \tfrac{65}{100} \qquad \overline{q} = \tfrac{35}{100}$$

$$z = \frac{\frac{30}{50} - \frac{35}{50}}{\sqrt{\frac{65}{100}\left(\frac{35}{100}\right)}\sqrt{\frac{1}{50} + \frac{1}{50}}} = -1.048$$

The value of z is between the critical values of -1.96 and 1.96; hence, we fail to reject the null hypothesis. Using the HP-21S, the reported P value is 0.147, which we must double because of the two-tailed test. The value would be at 29.4%, which is much too high.

Now we will do the hypothesis test using the chi-square test of homogeneity. We first need to find the expected values for each cell of the table, and then calculate X^2.

$$E_{1,1} = \frac{50(35)}{100} = 17.5 \qquad E_{1,2} = \frac{50(65)}{100} = 32.5$$

$$E_{2,1} = \frac{50(35)}{100} = 17.5 \qquad E_{2,2} = \frac{50(65)}{100} = 32.5$$

$$X^2 = \frac{(20-17.5)^2}{17.5} + \frac{(30-32.5)^2}{32.5} + \frac{(15-17.5)^2}{17.5} + \frac{(35-32.5)^2}{32.5}$$

$$X^2 = 0.357 + 0.192 + 0.357 + 0.192 = 1.098$$

The critical value of χ^2 at one degree of freedom is 3.84. The value of X^2 is less than the critical value; hence, we again fail to reject the null hypothesis. Using the HP-21S the reported P value for α is 29.4%. This is exactly the same value we found using the z test. The advantage of the z test is that one can test one tailed hypotheses. The advantage of the χ^2 test is that it can be generalized to more than two proportions.

Hot Spot #2– Reporting P Values For The χ^2 Distribution

Mann Sections 11.2 And 11.4

The table used to get critical values for the chi-square distribution is similar to the t table. The critical values listed for each value of α are found by first finding the degrees of freedom. However, there is a very important difference in the tables when we find the critical value for the left side of the curve.

The chi-square distribution unlike the z and t distributions, is not symmetrical. Because of the non-symmetry we list the critical values for each of the two tails of the curve. It is no longer possible to simply use a $\pm$ sign in front of the value we found for the right side of the curve. When we look at the chi-square table we find a heading that gives areas to the right of the critical value for both sides of the curve. A typical heading and row of critical values for $df = 9$ follows:

df	$\chi^2(0.99)$	$\chi^2(0.975)$	$\chi^2(0.95)$	$\chi^2(0.90)$	$\chi^2(0.10)$	$\chi^2(0.05)$	$\chi^2(0.01)$
9	2.088	2.700	3.325	4.168	14.684	16.919	21.666

The values in the table should make it very clear that we cannot simply use the $\pm$ symbol to get the value for the left side of the curve. For example, at $\alpha = 0.05$ the critical value of χ^2 for the right side is 16.919, but the critical value for the left side is 3.325.

When we do a hypothesis test using the χ^2 table we find the critical value or values consistent with whether we have a one-tailed or two-tailed test. We, of course, also need the significance level. It is when we decide to do a hypothesis test using a reported P value that we run into the same difficulty we had with reporting P values with the t distribution.

Before we look at how we handle the problem with the chi-square table, it is again important to understand that we can avoid the problem using a calculator like the HP-21S. Once we have the degrees of freedom entered into the calculator, we can find the area to the right of our observed value by pressing one key. For that matter, we can get critical values for a hypothesis test just as easily. However, the remainder of the discussion assumes that we will use the traditional χ^2 table to report the P value associated with a hypothesis test.

As an example, we will do a hypothesis test on H_0: $\sigma^2 = 1.25$ and H_a: $\sigma^2 > 1.25$ at $\alpha = 0.05$ with $n = 10$. The sample gave $s^2 = 2.20$. When we calculate the test statistic we get the following:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(9)(2.2)}{1.25} = 15.84$$

We then compare the calculated value to the table entries for $df = 9$. The calculated value is between 14.684 at $\alpha = 10\%$ and 16.919 at $\alpha = 5\%$. We would say that the hypothesis test was significant at a value between 5% and 10%. (Using the HP-21S, we find the exact value of $\alpha = 0.070$.)

Hot Spot #2 Sample Problem and Solution: Find the reported P value for the following hypothesis test on H_0: $\sigma^2 = 0.25$ and H_a: $\sigma^2 \neq 0.25$ with $s^2 = 0.5$ and $n = 10$.

Solution: $\chi^2 = \dfrac{(9)(0.5)}{0.25} = 18$ and $df = 10 - 1 = 9$.

18 is between 16.919 at $\frac{\alpha}{2} = 5\%$ and 19.023 at $\frac{\alpha}{2} = 2\frac{1}{2}\%$. The reported value of P is between 5% and 10%.

Sample Problem and Answer: Find the reported P value for a one-tailed hypothesis test on the left side of the curve if $\chi^2 = 5.22$ and $n = 15$.

Solution: We use $df = 15 - 1 = 14$.

5.22 is between 4.660 at $\alpha = 1\%$ and 5.629 at $\alpha = 2\frac{1}{2}\%$. The reported P value is between 1% and $2\frac{1}{2}\%$.

Sample Problem and Answer: Find the reported P value for a two-tailed hypothesis test if $\chi^2 = 50.475$ and $n = 30$.

Solution: We use $df = 30 - 1 = 29$.

50.475 is between 49.588 at $\frac{\alpha}{2} = 1\%$ and 52.336 at $\frac{\alpha}{2} = 0.5\%$. The reported P value is between 1% and 2%.

CHAPTER 11 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that your are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the difference between a test for goodness-of-fit and a test of independence?

2. What is the difference between a test for goodness-of-fit and a homogeneity test?

3. What is the difference between a test of independence and a homogeneity test?

4. What is the minimum expected frequency for any cell?

5. How do the row totals and column totals help determine if one is doing a test of independence or a test of homogeneity?

6. How does one find the expected frequencies for the cells for tests of independence and homogeneity tests?

7. What is the test statistic for a test of hypothesis about σ^2?

8. What is a contingency table?

9. How does one find degrees of freedom for the chi-square test?

10. How does the shape of the chi-square curve for $df = 1$ or $df = 2$ differ from the shape of the distribution for other values of degrees of freedom?